

Order Service API — Test Report

Manual endpoint verification via curl against http://localhost:8080 | Run on: macOS
(Nandans-MacBook-Air), working directory Assignment5

#	Endpoint	Method	Status	Result
1	/orders	GET	200	PASS
2	/orders	POST	201	PASS
3	/orders	GET	200	PASS
4	/orders/1	GET	200	PASS
5	/orders/1	GET	200	PASS
6	/orders/1/cancellation	POST	200	PASS
7	/orders/1	DELETE	204	PASS
8	/orders/999	GET	400	PASS
9	/orders	GET	401	PASS

1. List Orders — Initial State (Empty)

Method & Path	GET /orders
Request Headers	Authorization: Bearer demo
Status	200
Response Body	[]

Notes: Returns an empty JSON array before any orders have been created. Confirms the endpoint is reachable and authenticated correctly.

2. Create Order

Method & Path	POST /orders
Request Headers	Authorization: Bearer demo Content-Type: application/json Idempotency-Key: abc123
Request Body	{ "customerName": "Nandan", "totalAmount": 500 }
Status	201
Response Headers	Location: /orders/1 X-RateLimit-Limit: 100 X-RateLimit-Remaining: 99

Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>
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Notes: Order created successfully with HTTP 201. Response includes a Location header pointing to the new resource (/orders/1) and rate-limit headers. Idempotency-Key was supplied to guard against duplicate submissions.

3. List Orders — After Creation

Method & Path	GET /orders
Request Headers	Authorization: Bearer demo
Status	200
Response Body	<pre>[{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED", "createdAt": "2026-09-19T05:28:55.262595Z" }]</pre>

Notes: The previously created order is now present in the collection, including a server-generated createdAt timestamp.

4. Get Order by ID

Method & Path	GET /orders/1
Request Headers	Authorization: Bearer demo
Status	200
Response Headers	ETag: "1453530583" Cache-Control: max-age=60
Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>

Notes: Single-resource fetch succeeds and returns caching metadata (ETag, Cache-Control: max-age=60) suitable for conditional requests.

5. Get Order with Non-Matching If-None-Match

Method & Path	GET /orders/1
Request Headers	Authorization: Bearer demo If-None-Match: "12345"

Status	200
Response Headers	ETag: "483808708" Cache-Control: max-age=60
Response Body	{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }

Notes: Since the supplied If-None-Match value does not match the current ETag, the server returns 200 with the full body and a fresh ETag, rather than 304 Not Modified.

6. Cancel Order

Method & Path	POST /orders/1/cancellation
Request Headers	Authorization: Bearer demo
Status	200
Response Body	{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CANCELLED" }

Notes: Order status transitions from CREATED to CANCELLED via a dedicated cancellation sub-resource endpoint.

7. Delete Order

Method & Path	DELETE /orders/1
Request Headers	Authorization: Bearer demo
Status	204
Response Body	(empty)

Notes: Deletion succeeds with 204 No Content and an empty response body, as expected for a successful DELETE.

8. Get Non-Existent Order

Method & Path	GET /orders/999
Request Headers	Authorization: Bearer demo
Status	400
Response Body	{ "title": "Bad Request", "status": 400, "detail": "Order not found" }

Notes: Order 999 does not exist (and order 1 was just deleted). The API returns a problem-detail style error body. Note: semantically a 404 Not Found would be more conventional than 400 Bad Request for a missing resource — worth flagging to the backend team.

9. List Orders — Missing Authorization

Method & Path	GET /orders
Request Headers	(no Authorization header)
Status	401
Response Body	<pre>{ "timestamp": "2026-09-19T05:30:11.488+00:00", "status": 401, "error": "Unauthorized", "path": "/orders" }</pre>

Notes: Request without a Bearer token is correctly rejected with 401 Unauthorized and Spring Boot's default error payload shape.